

RILEY HACKETT

rileyhac@gmail.com

Ph: 0488 680 580

LinkedIn: [Riley Hackett](#)

[Engineering Portfolio](#)

Summary	Mechatronic Engineering graduate (pending conferral) from UNSW with extensive experience across many personal and academic projects in CAD-driven mechanical design, prototyping, and robotics-focused development. Complemented by university training in hardware–software integration for autonomous systems, with project work focused on robotics and integration across sensing, control, and implementation.	
Technical & Professional Skills	• Robotics: ROS2, robot kinematics, motion planning, localisation, hardware integration • Control & Estimation: PID control, state estimation, EKF, sensor fusion • Computer Vision: OpenCV (2D image/video processing), RGB-D & point-cloud processing (RANSAC, SVD), Fiducial markers • Programming: C++, Python, MATLAB, C, JavaScript, HTML, Git • Embedded Systems: Microcontrollers, sensors, actuators, I²C, UART, PCB assembly • CAD, CAM & Simulation: Fusion360, SolidWorks, COMSOL, ANSYS Granta • Professional Skills: Collaboration, analytical thinking, adaptability, attention to detail	
Education	Bachelor of Mechatronic Engineering (Honours) University of New South Wales Sydney, NSW	Feb 2022 – Aug 2026
	• Expected graduation: Dec 2026 Distinction average • Awarded UNSW’s “best and brightest” award for faculty performance • Honours thesis: Improving the effectiveness of virtual reality as a learning tool in robotics education Pending publication approval into the AAEE conference.	
	High School Certificate Hunter Valley Grammar School Ashtonfield, NSW	Feb 2016 – Nov 2021
	• Achieved 92.85 ATAR • Received the O’Grady Prize for excellence in STEM curricula • First Place Engineering Studies and Software Design & Development	
Recent Projects	Honours Thesis (86% HD) Improving the effectiveness of virtual reality as a learning tool in robotics education	Feb 2025 – Nov 2025
	• <i>Designed and implemented a VR simulation of the UR5e 6-DOF robotic manipulator, alleviating lab bottlenecks and improving the efficiency of robotics classes</i>	
	MTRN4231 – Mechatronics Capstone Project (91% HD) Automated robotic solution for fast-food burger production	Sep 2025 – Nov 2025
	• <i>Produced a client-focused automated burger assembly system using ROS2, 6-DOF path planning, end effector development, YOLO based computer vision, real-time visualisation and embedded programming.</i>	
	MTRN3100 – Micromouse Competition (85% HD) Autonomous maze solving wheeled robot (finished in the top 10%)	Jun 2025 – Aug 2025
	• <i>Designed, assembled, and programmed an Arduino-based autonomous micromouse, integrating sensor fusion (Time of Flight, IMU, encoders), control theory, computer vision, and path-planning algorithms</i>	
	DESN3000 – 3rd Year Design Project (94% HD) Autonomous gutter cleaning robot	Jun 2024 – Aug 2024
	• <i>Designed a conceptual autonomous gutter-cleaning robot to address fire-risk and maintenance challenges, using CAD for mechanical systems and proposing control and manufacturing strategies for practical implementation</i>	

RILEY HACKETT

rileyhac@gmail.com

Ph: 0488 680 580

LinkedIn: [Riley Hackett](#)

[Engineering Portfolio](#)

DESN2000 – 2nd Year Design Project (95% HD)

Hand cranked emergency UV water purifier

Sep 2023 –

Nov 2023

- Designed a conceptual hand-cranked UV water purifier to provide emergency water purification for hikers, using CAD to model mechanical components and designing the circuitry for practical, real-world functionality; currently used as a teaching example

Work Experience

Contactile – Engineering Intern

Cicada Innovations | Eveleigh Sydney, NSW

May 2026 –

Present

- Developing mechatronic cyclic testing rigs for tactile sensor validation and data collection.

VIPeople – Event Staff

Entertainment Quarter | Moore Park Sydney, NSW

Nov 2024 –

Feb 2025

- Front-of-House staff at Cirque du Soleil Sydney

QIB Group – Contract Consultant

Remote work

Oct 2023 –

Jun 2024

- Data processing for Insurance broking business

Floor Manager for Brick Cleaning Contractor

Newcastle TAFE (Steve Henderson Contractor) | Newcastle, NSW

Aug 2018 –

Dec 2021

- Disassemble walls and clean bricks built by TAFE bricklayer students
- Liaised with TAFE teaching staff & managed up to 5 employees delegating responsibilities

Student Tutor

Hunter Valley Grammar School | Ashtonfield, Newcastle NSW

Mar 2020 –

Dec 2020

Training & Certificates

NSW Food Handler Basics Training

NSW Food Authority | Online

Oct 2024

First Aid Certificate

Aspire Training & Development | Newcastle NSW

Jan 2024

Responsible Service of Alcohol

Aspire Training & Development | Newcastle, NSW

May 2022

Responsible Conduct of Gambling

Aspire Training & Development | Newcastle, NSW

May 2022

NSW Work Health & Safety General Construction White Card

Intercept Group Pty Ltd | Newcastle NSW

Feb 2020

Referees

Available upon request
